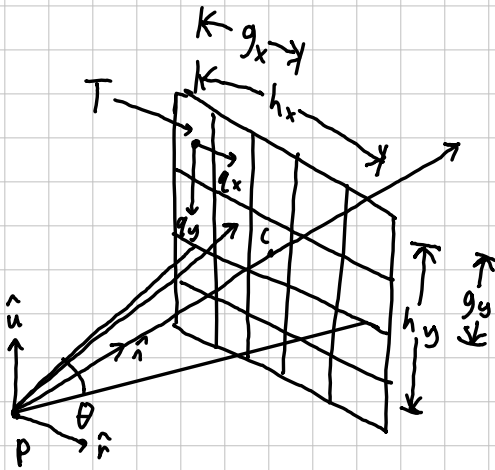


Raytracing?



camera axes

$$\hat{u} = \langle 0, 1, 0 \rangle$$

$$\hat{r} = \frac{\hat{n} \times \hat{u}}{\|\hat{n} \times \hat{u}\|}$$

$$\hat{n} = \hat{r} \times \hat{u}$$

viewport dimensions

$$\frac{h_x}{2} = opp, \quad \tan \frac{\theta}{2} = \frac{opp}{adj}, \quad adj = 1$$

$$\rightarrow g_x = \frac{h_x}{2} = \tan \frac{\theta}{2}, \quad \frac{h_y}{h_x} = \frac{g_y}{g_x} = \frac{ScreenHeight}{screenWidth}$$

$$\rightarrow g_y = g_x \frac{ScreenHeight}{screenWidth}$$

shifting vectors

$$\vec{q}_x = \frac{2g_x}{ScreenWidth} \hat{r}, \quad \vec{q}_y = \frac{2g_y}{ScreenHeight} \hat{u}$$

$$T = \hat{n} - g_x \hat{r} - g_y \hat{u}$$

actual spot on viewport

$$p_{ij} = T + \vec{q}_x i + \vec{q}_y j$$

$$\vec{r}_{ij} = \frac{p_{ij}}{\|p_{ij}\|}$$

finally p_o^v